

Predictive Logic Documentation Module

Module: PLDM — Predictive Logic Documentation Module

Parent Standard: PMES — Prediction Methodology Standard

Architecture: PGA™ — Prediction Governance Architecture

Category: Governance & Enforcement

Subcategory: Prediction Governance

Governed Space: Predictive Logic Documentation

Primary Governed Object: Documented Predictive Logic

Module Type: Core Module

Canonical Language: UCL — Universal Canonical Language

Protection: MIP® — Methodological Intellectual Property

1. Canonical Definition

The Predictive Logic Documentation Module (PLDM) governs the identification, documentation, traceability, coherence, and preservation of the material logic through which a defined Prediction Methodology transforms predictive inputs, conditions, intermediate states, methodological operations, judgments, or component outputs toward a predictive result.

PLDM establishes the documented reasoning and transformation logic connecting methodological structure to prediction formation.

A methodology may have identifiable components and structure while still failing to explain how those components methodologically lead to the prediction.

Defined Methodological Structure ≠ Documented Predictive Logic

2. Operational Role

PLDM is the third Core Module within PMES.

It receives the Defined Methodological Structure from MSDM and establishes the material predictive logic operating through that structure.

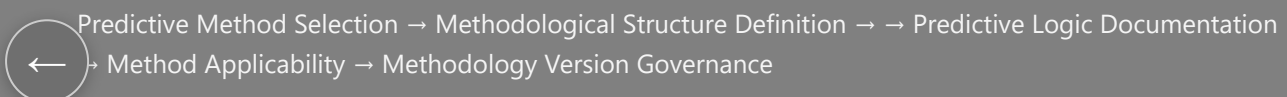
Its principal transition is:

Defined Methodological Structure → Documented Predictive Logic

PLDM governs how the methodology is intended to move from predictive conditions toward predictive output.

It does not determine the actual Prediction Basis, establish predictive assumptions, or form the specific prediction.

3. Internal Module Flow



Current Position: Predictive Logic Documentation

The documented predictive logic proceeds into:

MAPM — Method Applicability Module

4. Operational Space

PLDM operates within the Predictive Logic Documentation space.

It governs:

predictive logic identification; transformation logic; decision logic; weighting logic; aggregation logic; conditional logic; judgment logic; model-interaction logic; intermediate predictive states;

logic dependencies; logic traceability; logic coherence; logic documentation; logic change.

PLDM governs governance-relevant predictive logic.

It does not require disclosure of every technical operation performed by the methodology.

5. Operational Scope

Predictive logic may be:

statistical; mathematical; algorithmic; computational; rule-based; probabilistic; analytical; judgment-based; market-derived; aggregation-based; simulation-supported; AI-mediated; hybrid; another sufficiently defined form of predictive reasoning or transformation.

The form of the logic does not determine whether it can be governed.

6. Predictive Logic Identification

PLDM must identify the material logic by which the Prediction Methodology moves from its governed inputs and conditions toward a predictive result.

The governing question is:

By what material logic does this methodology transform what it receives into what it predicts?

Naming a model, algorithm, expert, market, or forecasting technique does not necessarily answer this question.

Method Identification $\neq$ Predictive Logic Identification

7. Transformation Logic

Where predictive information is transformed between methodological stages, the material transformation logic must remain identifiable.

Transformation may include:

normalization; classification; filtering; scoring; estimation; extrapolation; adjustment; synthesis; weighting; aggregation; interpretation; other prediction-relevant transformations.

PLDM governs the logic of the transformation.

It does not govern whether the underlying information itself is sufficient.

8. Input-to-Output Logic

A governed Prediction Methodology must preserve a sufficiently intelligible relationship between what enters the methodology and what emerges from it.

This does not require a simplistic one-to-one explanation.

Complex predictive systems may contain multiple intermediate transformations.

The requirement is that material methodological transitions remain governable.

Input Presence $\neq$ Explained Predictive Output

9. Intermediate Predictive States

Where a methodology produces material intermediate states before generating its final predictive output, those states must remain identifiable where necessary to understand predictive logic.

Intermediate states may include:

estimates; scores; classifications; scenario weights; model outputs; confidence-related indicators; expert adjustments; aggregated signals; other prediction-relevant states.

PLDM does not automatically classify every internal computational state as governance-relevant.

Only materially significant states require governance visibility.

10. Decision Logic

Where a methodology contains decision points that determine which predictive pathway, model, rule, component, or output proceeds, the material decision logic must remain identifiable.

Examples may include:

If Condition A $\rightarrow$ Method Path A

If Condition B $\rightarrow$ Method Path B

or selection among multiple model outputs.

An undocumented decision point can materially alter a prediction while remaining invisible in the represented methodology.

Visible Components $\neq$ Visible Decision Logic

11. Conditional Logic

Predictive methodologies may operate differently under different conditions.

Where conditional logic materially changes methodology behavior, the conditions and resulting methodological pathways must remain identifiable.

PLDM governs methodological conditionality.

The truth or legitimacy of predictive assumptions themselves belongs to PAS.

Methodological Condition $\neq$ Predictive Assumption

12. Weighting Logic

Where multiple signals, variables, models, expert judgments, or methodological components receive different influence within the predictive process, material weighting logic must remain identifiable.

The existence of multiple inputs does not imply equal methodological influence.

Multiple Inputs $\neq$ Equal Predictive Weight

This distinction becomes critical in ensembles, expert panels, hybrid systems, and market-supported methodologies.

13. Aggregation Logic

Where multiple methodological outputs are combined, PLDM governs the logic by which aggregation contributes to the resulting prediction.

Aggregation may involve:

averaging; weighted combination; voting; ranking; consensus mechanisms; model selection; market mechanisms; expert synthesis; another documented aggregation process.

Multiple Outputs ≠ Governed Aggregation

Where multiple independent predictive sources form a collective prediction, CPGS provides additional governance.

14. Human Judgment Logic

Where human judgment materially contributes to prediction formation, PLDM must document its methodological role sufficiently to prevent the human contribution from becoming an invisible transformation.

Human judgment may include:

adjustment; interpretation; weighting; exclusion; escalation; synthesis; override; selection.

PLDM does not require human judgment to become algorithmic.

It requires its material methodological function to remain identifiable.

Human Judgment ≠ Methodological Black Box by Default

15. AI Predictive Logic

Where AI participates in Prediction Methodology, PLDM must establish the governance-relevant logic of its predictive role to the extent necessary to understand how AI output contributes to the prediction.

This does not require complete reconstruction of every internal model parameter or computation.

A technically opaque model may still participate in a governed methodology where its role, inputs, outputs, transformations, limitations, and surrounding methodological logic are sufficiently governed.

Technical Opacity ≠ Automatic Methodological Ungovernability

But:

Model Output Alone ≠ Documented Predictive Logic

16. Prediction Market Logic

Where prediction-market outputs participate in methodology, PLDM must preserve the methodological logic by which the market signal is interpreted or transformed into predictive output.

A market price may be converted into a probability-like representation or incorporated into another predictive process.

That transformation must remain identifiable.

Market Price ≠ Governed Probability Estimate

The collective composition and influence structure behind the market remain governed elsewhere in PGA™ where applicable.

17. Simulation-Supported Logic

Where simulation contributes to prediction, PLDM must identify the methodological logic by which simulation outputs are used to support a predictive conclusion.

A simulated scenario may demonstrate possibility, pathway, sensitivity, or conditional behavior without independently establishing probability.

Simulated Outcome $\neq$ Predicted Outcome

Simulation Frequency $\neq$ Governed Probability by Default

SIMULOS® retains governance ownership of simulation itself.

18. Logic Dependencies

Predictive logic may depend upon:

particular methodological components; model configurations; external systems; thresholds; transformations;

human judgments; aggregation mechanisms; other methodological conditions.

Material dependencies must remain identifiable where their change could alter the predictive logic.

19. Logic Coherence

The documented predictive logic must be coherent with the methodological structure established by MSDM.

A documented methodology must not claim one predictive logic while operationally applying another.

Where material divergence exists, the methodology requires clarification.

Documented Logic $\neq$ Operational Logic unless they materially correspond

20. Logic Traceability

Material predictive transformations must remain sufficiently traceable to allow the methodology behind a prediction to be reconstructed at the governance-relevant level.

Traceability does not require exposure of every computational operation.

It requires preservation of the methodological path connecting relevant inputs, transformations, decisions, and outputs.

Technical Exhaustiveness $\neq$ Predictive Logic Traceability

21. Logic Explainability

Predictive logic must be explainable to the degree necessary for governance.

Explainability may vary depending upon methodological complexity and system type.

PLDM does not require every predictive mechanism to become intuitively understandable to every person.

It requires sufficient explanation to establish what governance-relevant transformations occur and how they contribute to predictive output.

Explainability $\neq$ Simplification

22. Logic Documentation Sufficiency

Documentation must capture the material logic necessary to distinguish the actual methodology from an undocumented or retrospectively invented explanation.

Sufficiency depends upon the predictive context.

The standard does not equate longer documentation with stronger governance.

More Documentation $\neq$ Better Predictive Logic Governance

The relevant test is whether the material predictive logic can be understood and traced.

23. Logic and Prediction Basis Separation

PLDM governs what the methodology does with predictive inputs.

It does not determine which information, data, observations, evidence references, or signals legitimately constitute the Prediction Basis.

That belongs to PBAS.

Predictive Logic $\neq$ Prediction Basis

A fully documented transformation does not establish that the information being transformed is sufficient or appropriate.

24. Logic and Assumption Separation

Predictive logic may operate under assumptions.

PLDM may identify where assumptions enter or affect methodological logic, but PAS governs assumption identification, dependency, sensitivity, materiality, and change.

Logic Dependency on an Assumption $\neq$ Assumption Governance

25. Logic and Prediction Formation Separation

PLDM documents the methodology capable of producing predictions.

PFOS governs formation of a specific governed prediction using the applicable methodology, basis, assumptions, and conditions.

Therefore:

Documented Predictive Logic $\neq$ Formed Prediction

The same predictive logic may support multiple predictive instances.

26. Logic Change

Predictive logic may change as methodology evolves.

Material changes may include:

altered transformations; changed thresholds; changed weighting; changed decision rules; changed aggregation; changed human intervention; changed model interaction; changed conditional pathways.

Material logic changes must remain identifiable.

Whether they constitute a new methodology version is governed by MVGM.

Changed Predictive Logic $\neq$ Same Methodology by Default

27. Predictive Logic Sufficiency

Predictive logic is sufficiently documented when:

material transformations are identifiable; relevant input-to-output relationships are represented; material intermediate states are visible where necessary;

decision and conditional logic are identifiable; material weighting and aggregation logic are documented; human or machine transformations are represented where material; dependencies remain identifiable; documented logic corresponds with methodological structure; material logic can be traced; material logic changes remain detectable.

Logic sufficiency does not establish Method Applicability or predictive correctness.

28. Governance Requirements

PLDM requires that:

1. material predictive logic be explicitly identifiable; 2. material transformations remain documented; 3. relevant input-to-output relationships remain traceable; 4. material intermediate states be identifiable where necessary; 5. decision logic remain visible where it can alter predictive output; 6. conditional logic remain identifiable where material; 7. weighting and aggregation logic remain documented where applicable; 8. material human judgment not remain an invisible transformation; 9. governance-relevant AI logic be sufficiently represented; 10. documented logic correspond with defined methodological structure; 11. material dependencies remain identifiable; 12. material logic changes remain traceable.

29. Governance Logic

PLDM applies the following operational sequence:

Defined Methodological Structure

↓

Material Predictive Logic Identified

↓

Transformations Documented

↓

Intermediate States Identified where material

↓

Decision / Conditional Logic Documented

↓

Weighting / Aggregation Logic Documented where applicable

↓

Human / Machine Logic Documented where applicable

↓

Logic Dependencies Identified

↓

Logic Coherence Assessment

↓

Logic Traceability Assessment

↓

Documented Predictive Logic

↓

MAPM — Method Applicability Module

Where logic sufficiency is not established:

Predictive Logic Governance Gap → Clarification / Documentation → Reassessment

30. Minimum Implementation Framework

A minimum implementation of PLDM must establish:

A. Methodological Structure Reference

Reference to the structure governed under MSDM.

B. Predictive Logic Statement

A governance-level description of how the methodology progresses toward predictive output.

C. Material Transformations

The transformations capable of materially affecting predictive meaning.

D. Intermediate States

Governance-relevant intermediate predictive states where applicable.

E. Decision Logic

Material decision points affecting methodological pathways.

F. Conditional Logic

Conditions that materially alter methodology operation.

G. Weighting Logic

Material weighting relationships where applicable.

H. Aggregation Logic

The logic governing combination of methodological outputs where applicable.

I. Human / Machine Logic

Material human or machine transformations affecting predictive output.

J. Logic Dependencies

Dependencies capable of materially changing predictive logic.

K. Logic Traceability State

Whether the material methodological path can be reconstructed.

L. Logic State

At minimum:

Undocumented → Partial → Clarification Required → Documented → Changed → Superseded

Only sufficiently Documented predictive logic proceeds as the governed logic into Method Applicability.

31. Governance Outputs

PLDM may produce:

Documented Predictive Logic; predictive logic statement; transformation map; decision-logic reference; conditional-logic reference; weighting logic; aggregation logic; intermediate-state references; human-machine logic reference; logic dependencies; traceability state; logic documentation state; unresolved logic requirements.

The primary governed output is:

Documented Predictive Logic

This becomes a principal input into:

MAPM — Method Applicability Module

32. Critical Distinctions

Defined Methodological Structure ≠ Documented Predictive Logic
Method Identification ≠ Predictive Logic Identification
Input Presence ≠ Explained Predictive Output
Visible Components ≠ Visible Decision Logic
Methodological Condition ≠ Predictive Assumption
Multiple Inputs ≠ Equal Predictive Weight
Multiple Outputs ≠ Governed Aggregation
Human Judgment ≠ Methodological Black Box by Default
Model Output Alone ≠ Documented Predictive Logic
Market Price ≠ Governed Probability Estimate
Simulated Outcome ≠ Predicted Outcome
Simulation Frequency ≠ Governed Probability by Default
Documented Logic ≠ Operational Logic unless they materially correspond
Technical Exhaustiveness ≠ Predictive Logic Traceability
Predictive Logic ≠ Prediction Basis
Documented Predictive Logic ≠ Formed Prediction

33. Enterprise Application

PLDM creates a governance layer between knowing which systems and methods are involved in prediction and understanding how those systems and methods actually produce predictive meaning.

This becomes critical where predictive systems contain complex chains of transformations, AI models, expert adjustments, aggregation mechanisms, simulation outputs, external services, or market-derived signals.

Without documented predictive logic, an organization may possess detailed technical infrastructure while remaining unable to explain the methodological path behind a consequential prediction.

34. Use Case 1 — A 92% Probability with No Traceable Path

A decision system presents a prediction:

"92% probability of operational failure within 12 months."

The organization knows which AI model produced the initial analysis and which data systems were connected.

However, it cannot establish how the system moved from those inputs to 92%.

It is unclear which signals were weighted, which were excluded, whether intermediate model outputs were transformed, whether human adjustment occurred, or whether the displayed percentage was generated directly by the model or through a later calculation.

The governance question is:

Can a probability be treated as a governed predictive output when the methodological path producing that probability cannot be reconstructed?

PLDM requires the material predictive logic behind the transformation to remain documented and traceable.

Numerical Precision ≠ Methodological Traceability

35. Use Case 2 — Expert Override Hidden Behind an AI Forecast

An AI forecasting system generates a 55% probability for an outcome.

A senior expert considers the result too low and changes the published prediction to 75% based on professional judgment.

The organization continues to describe the final result as an “AI prediction.”

No governed record explains the adjustment or the logic by which 55% became 75%.

The governance question is:

What methodology actually produced the 75% prediction—the AI model, the expert, or a hybrid methodology whose logic was never documented?

PLDM does not prohibit the expert adjustment.

It requires the transformation logic to become part of the governed Prediction Methodology when it materially changes the predictive output.

An undocumented adjustment is not documented predictive logic merely because the person making it is an expert.

36. Canonical Boundary

PLDM governs the identification, documentation, traceability, coherence, and preservation of the material predictive logic connecting methodological structure, transformations, decision points, conditional pathways, weighting, aggregation, and human-machine contributions toward predictive output. It does not govern Prediction Basis sufficiency, predictive assumptions, formation of the specific prediction, probability representation, feasibility, final Method Applicability, methodology version governance, Operational Reality, or predictive performance.

PLDM establishes how the methodology methodologically moves toward a prediction .

It does not establish whether the resulting prediction is correct .

Canonical Closing Statement

Knowing what systems produced a prediction is not enough if the logic connecting those systems to the predictive result remains unknown.

PLDM makes the methodological path behind the prediction governable before the output is allowed to stand in place of its reasoning.

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contact@originopenfoundation.org